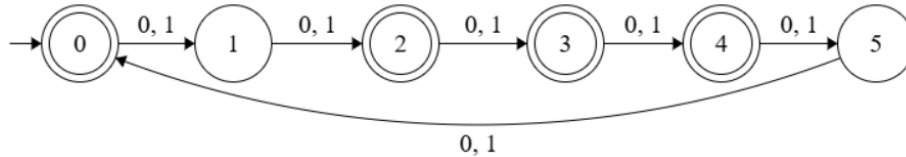


BRAC UNIVERSITY
CSE331 : Automata and Computability
Assignment 1

1. Draw the state diagram of a DFA for the following regular languages:

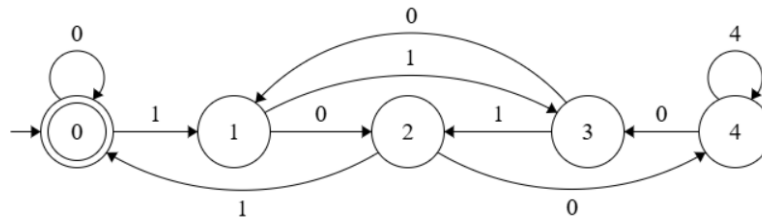
A. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the length of } w \text{ is a multiple of 2 or 3}\}$, where $\Sigma = \{0, 1\}$. (use 6 states)



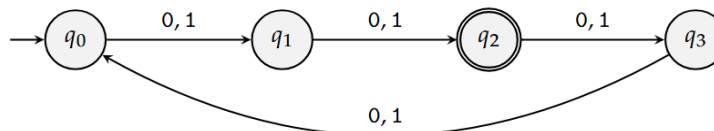
B. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the sum of the symbols of } w \text{ is a multiple of 3}\}$, where $\Sigma = \{0, 1, 2\}$.

See example 1.13 from the book.

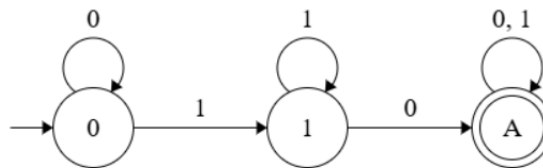
C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the decimal equivalent of } w \text{ is a multiple of 5}\}$, where $\Sigma = \{0, 1\}$.



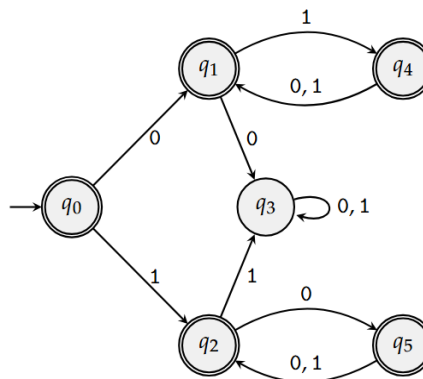
D. $L(M) \rightarrow \{w \in \Sigma^* \mid |w| \% 4 = 2\}$



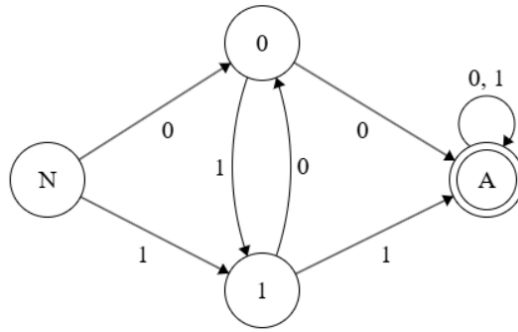
E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ is any string not in } 0^*1^*\}$, where $\Sigma = \{0, 1\}$.



F. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{Every even position letter in } w \text{ is different from the first letter of } w\}$



G. $L(M) \rightarrow (L_1 \cap L_2)'$



2. Write the RE for the following regular languages:

A. $L(M) \rightarrow \{w \in \Sigma^* \mid |w| \% 3 \neq 1\}$

$(\Sigma\Sigma\Sigma)^*(\varepsilon \mid \Sigma\Sigma)$

B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ starts and ends with the same symbol}\}$, where $\Sigma = \{0, 1\}$.

$0\Sigma^*0 \mid 1\Sigma^*1 \mid 0 \mid 1$

C. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of } 01 \text{ and } 10\}$, where $\Sigma = \{0, 1\}$.

$0\Sigma^*0 \mid 1\Sigma^*1 \mid 0 \mid 1 \mid \varepsilon$

D. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains at most two } 11\}$

$(0 \mid 10)^*(11 \mid \varepsilon)(0 \mid 01)^* \mid (0 \mid 10)^*(11)(0 \mid 01)^*0(11)(0 \mid 01)^* \mid (0 \mid 10)^*(111)(0 \mid 01)^* \mid 1$

E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ does not contain } 101\}$, where $\Sigma = \{0, 1\}$.

$(1|000^*)^*|\varepsilon|\Sigma|\Sigma\Sigma$

F. $L(M) \rightarrow L_1'$

$(\Sigma\Sigma\Sigma)^*\Sigma\Sigma 0\Sigma^*$

G. $L(M) \rightarrow L_1 \cap L_2$

$(001)^*00 \mid \varepsilon \mid 0 \mid 00$